The Witten Effect and Cohomology

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Abstract

This expository paper introduces the Witten effect, first as a direct result of quantizing electromagnetism, and eventually in the modern language of differential cohomology.

1 Introduction

The Dirac quantization condition is a well-known result: the quantum mechanics of a pure electric charge q and a pure magnetic monopole g is consistent only if $qg = 2\pi n$, with $n \in \mathbb{Z}$ [Dir31]. Two dyons, i.e. particles that carry both electric and magnetic charge, of electric and magnetic charges (q, g), (q', g'), can only coexist if $qg' - q'g = 2\pi n$ [Sch66] [Zwa68]. This is the Zwanziger-Schwinger quantization condition. Assume the electron charge e is the electric charge of smallest magnitude carried by any dynamical particle in the theory. Then by the Zwanziger-Schwinger condition, a dyon of charge (q, g) must satisfy $eg - 0 \cdot q = 2\pi n$, i.e. $g = \frac{2\pi n}{e}$. That is, any magnetic charge is quantized in units of $\frac{2\pi}{e}$; in particular, the minimum nonzero magnetic charge allowed by the quantization condition is $g_{\text{min}} = \frac{2\pi}{e}$, assuming the spectrum contains a monopole saturating this bound. For two dyons of charges (q, g_{min}), (q', g_{min}), Zwanziger-Schwinger gives $q - q' = ne$. Thus for fixed g_{min}, all allowed charges differ by integer multiples of e; there is then some offset q_0 such that $q = q_0 + ne$. Under CP, $(q, g) \mapsto (-q, g)$ because the electric charge changes sign, while magnetic charge is CP even. Therefore, in a CP-invariant theory, if (q, g) exists, then $(-q, g)$ must also exist. But these two dyons have the same magnetic charge, so their electric charges must differ by an integer multiple of e; $q - (-q) = ne$, hence $q = \frac{n}{2}e$. Modulo shifts by integer multiples of e, there are then only two possible CP-invariant offsets: $q_0 = 0$ or $q_0 = \frac{e}{2}$. Equivalently, the electric charges in the minimal monopole sector are either $q \in e\mathbb{Z}$ or $q \in e\left(\mathbb{Z} + \frac{1}{2}\right)$.

This gets modified in the presence of a CP-violating term, and the electric charge of the dyon will depend explicitly on some θ angle measuring the extent of CP violation; this is called the Witten effect [FO98]. In elementary treatments ([FO98], [Sch13]), the so-called "θ term" $S_\theta = \frac{\theta}{8\pi^2}\int\langle F \wedge F\rangle$ is often introduced somewhat ad hoc as an additional Lorentz- and gauge-invariant interaction proportional to $\vec{E} \cdot \vec{B}$. Since this density is locally a total derivative, it does not contribute to the classical Maxwell equations in topologically trivial sectors. In the presence of magnetic monopoles, i.e. particles of charge $(0, g)$ when $\theta = 0$, the gauge field is not globally described as a single smooth potential. As a result, a monopole of magnetic charge $g = \frac{2\pi m}{e}$ for some $m \in \mathbb{Z}$ acquires electric charge an electric charge shift of $\frac{e\theta}{2\pi}m$. So θ displaces the electric charge lattice, with (most commonly) the periodicity $\theta \sim \theta + 2\pi$ realized by a relabeling of states $n \mapsto n + m$.

More generally, the Witten effect is the modification of the electric charge lattice of magnetically charged states or defects in the presence of topological terms in the action. Since both ingredients in this statement—magnetic charge and the θ term—are global in nature, a cohomological formulation is especially natural. Magnetic charge is measured by the flux of F through a surrounding two-sphere, equivalently by an integral cohomology class, while the θ term is determined by the cup-product pairing of such classes on spacetime. The Witten effect can therefore be viewed as a topological shift of the electric charge lattice induced by this pairing.

The paper is structured as follows. Section 2 obtains the Witten effect for a $U(1)$ Yang-Mills theory with the "topological Yang-Mills" term (i.e. θ term). Section 3 shows how the magnetic charge of a monopole can be computed using integral cohomology, and derives a "topological Dirac quantization" for a general $U(1)$ line bundle. Section 4 introduces the 't Hooft-Polyakov monopole and computes its magnetic charge using the result of section 3. Section 5 notes that the periodicity of θ depends on whether or not the manifold is spin. Finally, section 6 describes the ordinary $U(1)$ Witten effect in modern language, and we conclude with an eye towards higher generalizations.

2 The canonical Witten effect

Consider a Yang-Mills theory, i.e. a gauge theory on a 4-dimensional pseudo-Riemannian manifold M whose field is represented by a vector bundle with connection, and whose action functional is

$$S : \nabla \mapsto -\frac{1}{2\alpha^2}\int_M \langle F_\nabla \wedge \star F_\nabla \rangle + \frac{\theta}{8\pi^2}\int_M \langle F_\nabla \wedge F_\nabla \rangle \tag{1}$$

where F_∇ is the field strength (given locally by the curvature $\mathfrak{u}(n)$-valued differential form on M), $\star$ is the Hodge star operator of the metric, $\frac{1}{\alpha^2}$ is the Yang-Mills coupling constant, and θ is the theta angle, and $\langle -, - \rangle : \mathfrak{g} \otimes \mathfrak{g} \to \mathbb{R}$ is an invariant bilinear form which serves to contract gauge indices. Note that the second term of S in (1) vanishes in CP-invariant Yang-Mills theory. $\langle F_\nabla \wedge F_\nabla \rangle$ is the Chern-Weil representative of a degree-four characteristic class of the principal gauge bundle. Unlike $\langle F_\nabla \wedge \star F_\nabla \rangle$, it does not use the metric; on a closed four-manifold its integral depends only on the topology of the bundle.

In ordinary $U(1)$ Yang-Mills theory,

$$\mathcal{L} = \frac{1}{2\alpha^2}(|\vec{E}|^2 - |\vec{B}|^2) + \frac{\theta}{4\pi^2}\vec{E} \cdot \vec{B}$$

We normalize[1] the $U(1)$ gauge field so that a unit-charged matter field couples as $D_\mu\psi = (\partial_\mu - iA_\mu)\psi$. Thus a gauge transformation by parameter β acts as $\psi \mapsto e^{i\beta}\psi$, and because the gauge group is compact $U(1)$, $\beta \sim \beta + 2\pi$. More generally, a state of canonical charge q transforms as $|\Psi\rangle \mapsto e^{iq\beta}|\Psi\rangle$. Single-valuedness under the identity transformation $\beta = 2\pi$ requires $e^{2\pi i q} = 1$, so $q \in \mathbb{Z}$. The quantity q is precisely the canonical charge generated by Gauss's law $q = \int_{S^2} \vec{D} \cdot d\vec{S}$. The canonical electric displacement for our Lagrangian is

$$\vec{D} = \frac{\partial \mathcal{L}}{\partial \vec{E}} = \frac{1}{\alpha^2}\vec{E} + \frac{\theta}{4\pi^2}\vec{B}$$

By Gauss' law, the canonical charge measured on any closed two-sphere S^2 lying outside the monopole core is

$$q = \int_{S^2} \vec{D} \cdot d\vec{S} = \frac{1}{\alpha^2}\int_{S^2} \vec{E} \cdot d\vec{S} + \frac{\theta}{4\pi^2}\int_{S^2} \vec{B} \cdot d\vec{S}$$

If S^2 links the monopole worldline once, then magnetic flux quantization gives $\int_{S^2} \vec{B} \cdot d\vec{S} = 2\pi m$ for some $m \in \mathbb{Z}$. Hence $q = \frac{1}{\alpha^2}\int_{S^2} \vec{E} \cdot d\vec{S} + \frac{\theta}{2\pi} m$. Since $q \in \mathbb{Z}$,

$$q_{\text{electric}} = \frac{1}{\alpha^2}\int_{S^2} \vec{E} \cdot d\vec{S} = n - \frac{\theta}{2\pi} m$$

for some $n \in \mathbb{Z}$, which is precisely the Witten effect shift.

3 Magnetic charge as integral cohomology

We first isolate the topological content of magnetic charge in the setting of a $U(1)$ gauge field on a two-sphere surrounding the monopole. When the long-distance abelian gauge field is a connection on a complex line bundle, its magnetic flux is the first Chern number of that line bundle.

Let $L \to S^2$ be a complex line bundle with connection. Cover S^2 by northern and southern patches U_N, U_S; on each patch, the connection is represented by an ordinary 1-form $A_N \in \Omega^1(U_N), A_S \in \Omega^1(U_S)$. On the overlap $U_N \cap U_S$, they are related by a gauge transformation $A_N = A_S + i\phi^{-1}d\phi$, where $\phi : U_N \cap U_S \to U(1)$. The local curvatures agree on the overlap ($F_N = dA_N = dA_S = F_S$), so they glue to a globally defined 2-form F on S^2. Therefore $\int_{S^2} F = \int_{U_N} dA_N + \int_{U_S} dA_S$. By Stokes' theorem, and using the induced orientations on the two hemispheres, this reduces to an integral over the equator, $\int_{S^1_{eq}}(A_N - A_S)$. In particular, $\frac{1}{2\pi}\int_{S^2} F = \frac{i}{2\pi}\int_{S^1_{eq}} \phi^{-1}d\phi$. But $\frac{i}{2\pi}\int_{S^1} \phi^{-1}d\phi$ is the winding number of the transition function $\phi : S^1 \to U(1)$, so it is an integer; in particular, it is the first Chern number of the line bundle:

$$\frac{1}{2\pi}\int_{S^2} F = \langle c_1(L), [S^2]\rangle \in \mathbb{Z}$$

[1] For the remainder of this paper, we set the electron charge e to be 1.

Equivalently,

$$\left[\frac{F}{2\pi}\right] = c_1(L) \otimes 1 \in H^2(S^2; \mathbb{R})$$

where $\left[\frac{F}{2\pi}\right]$ is the image of the first Chern class $c_1(L) \in H^2(S^2; \mathbb{Z})$ under the natural map $H^2(S^2; \mathbb{Z}) \to H^2(S^2; \mathbb{R})$. In the monopole problem, S^2 is taken to be a sphere linking the monopole worldline, and F is the curvature of the residual long-distance $U(1)$ connection. The magnetic charge is therefore the Chern number of this line bundle:

$$m = \frac{1}{2\pi} \int_{S^2} F \in \mathbb{Z}$$

This is the "topological" Dirac quantization due to [WY76] and by a construction of [Alv85].

4 The 't Hooft-Polyakov monopole

We now explain how the preceding $U(1)$ line-bundle picture arises from a smooth nonabelian monopole. Consider an $SO(3)$ Yang-Mills theory defined on a 4-dimensional Lorentzian spacetime M, coupled to an adjoint Higgs field $\Phi \in \Omega^0(M, \mathfrak{so}(3))$. The curvature is $F_A = dA + \frac{1}{2}[A \wedge A]$ and the covariant derivative of the Higgs field is $D_A\Phi = d\Phi + [A, \Phi]$. The bosonic energy density on a spatial slice Σ contains the terms

$$\mathcal{H} = \frac{1}{2}|E|^2 + \frac{1}{2}|B|^2 + \frac{1}{2}|\Pi|^2 + \frac{1}{2}|D_a\Phi|^2 + V(\Phi)$$

where $V(\Phi) = \frac{\lambda}{4}(\langle\Phi, \Phi\rangle - \nu^2)^2$ for some $\lambda \geq 0$, and $\langle\cdot, \cdot\rangle$ is an invariant inner product on $\mathfrak{so}(3)$. The pointwise vacuum manifold is $\mathcal{V} = \{\Phi \in \mathfrak{so}(3) : \langle\Phi, \Phi\rangle = \nu^2\} \cong S^2$. Finite-energy requires the Higgs field to approach the vacuum manifold at spatial infinity. Hence, as $r \to \infty$, we have $|\Phi| \to \nu$ and $D_A\Phi \to 0$ up to the residual long-range abelian field. Choosing a reference vacuum $\Phi_0 = \nu T_3$ (with T_3 a reference generator of $\mathfrak{so}(3)$), its stabilizer is Stab$(\Phi_0) \cong SO(2)$, so the symmetry breaking pattern is $SO(3) \to SO(2) \cong U(1)$. Thus the low-energy gauge field at infinity is an abelian gauge field valued in the line subalgebra stabilizing the Higgs direction. Equivalently, after choosing local gauges in which the Higgs direction is fixed, the surviving field is a connection on the residual $SO(2) \cong U(1)$ bundle over S^2_∞. The Higgs field need not approach the same vacuum direction at every point of S^2_∞; instead, the normalized Higgs field $\widehat{\Phi} = \frac{\Phi}{|\Phi|}$ defines a map $\widehat{\Phi} : S^2_\infty \to \mathcal{V} \cong S^2$, whose homotopy class $[\widehat{\Phi}] \in \pi_2(S^2) \cong \mathbb{Z}$ is the monopole number m. A 't Hooft-Polyakov monopole is a finite-energy configuration in which the Higgs field winds nontrivially at infinity, i.e. one for which $m \neq 0$. Although the configuration is nonsingular in the full nonabelian theory, at long distances it appears as a magnetic monopole for the surviving

$U(1)$ gauge field. A gauge-invariant representative of the electromagnetic field strength is the tensor [tHo74]

$$F_{em} = \langle \widehat{\Phi}, F_A \rangle - \langle \widehat{\Phi}, D_A \widehat{\Phi} \wedge D_A \widehat{\Phi} \rangle$$

up to the normalization convention for the generators of $\mathfrak{so}(3)$. On a local patch of S^2_∞, choose a gauge in which $\widehat{\Phi} = T_3$. In this gauge, the condition $D_A \widehat{\Phi} = 0$ at infinity implies that the surviving component of A lies in $\mathfrak{so}(2) = \mathbb{R}T_3$, and F_{em} reduces to the curvature of this unbroken $SO(2) \cong U(1)$ connection. Our description of the topological Dirac quantization was for a general $U(1)$ line bundle over S^2. For the 't Hooft-Polyakov monopole, the residual $SO(2)$-bundle arising from an $SO(3)$ monopole is constrained to come from the fibration $SO(2) \to SO(3) \to S^2$. Recall that $\pi_1(SO(2)) \cong \mathbb{Z}$ and $\pi_1(SO(3)) \cong \mathbb{Z}/2$. From the long exact sequence

$$0 \to \pi_2(S^2) \xrightarrow{\partial} \pi_1(SO(2)) \cong\to \pi_1(SO(3)) \to 0$$

we have $\operatorname{im}\partial = 2\mathbb{Z}$. Hence a Higgs field of degree m gives $SO(2)$ transition of winding $2m$; equivalently, $\frac{1}{2\pi}\int_{S^2_\infty} F_{SO(2)} = 2m$. Thus, in this $SO(3)$ normalization, the Chern number of the residual $SO(2)$ bundle is twice the Higgs winding number, following from the normalization of the embedding $SO(2) \subset SO(3)$.

5 Aside: Periodicity of θ

On a general closed oriented four-manifold X, the cup product defines an integral symmetric pairing $B(a, b) = \int_X a \smile b$ for $a, b \in H^2(X;\mathbb{Z})$. For a $U(1)$ field strength with $a = \left[\frac{F}{2\pi}\right]$, the abelian θ-term is governed by $\frac{1}{8\pi^2}\int_X F \wedge F = \frac{1}{2}\int_X \frac{F}{2\pi} \wedge \frac{F}{2\pi}$, which represents $Q(a) = \frac{1}{2}\int_X a \smile a$. Thus $S_\theta = \theta Q(a)$. If X is spin, then by the Wu formula (cf. [May99]) the intersection form is even, i.e. $\int_X a \smile a \in 2\mathbb{Z}$ for every $a \in H^2(X;\mathbb{Z})$. Hence $Q(a) \in \mathbb{Z}$. The weight in the path integral $e^{iS_\theta} = e^{i\theta Q(a)}$ is therefore invariant under $\theta \sim \theta + 2\pi$. On a general oriented four-manifold, however, the intersection form need not be even, so $Q(a)$ can be half-integral. The naive expression above then generally has 4π, rather than 2π, periodicity in θ. Indeed, this agrees with the "statistical Witten effect" of [MKF13]; magnetic monopoles in bosonic topological insulators have bosonic statistics at $\theta = 0$, while they have fermionic statistics at $\theta = 2\pi$, and become bosons once again at $\theta = 4\pi$. On the other hand, monopoles in fermionic insulators have bosonic statistics at both $\theta = 0$ and $\theta = 2\pi$.

6 Differential cohomology

In classical electromagnetism on Minkowski spacetime M, the field strength 2-form F is related to the electric and magnetic current three-forms j_E, j_B by Maxwell's equations $dF = j_B$, $d \star F = j_E$. We assume that on any spacelike slice j_E, j_B have compact support (or more generally satisfy some integrability condition). The integral of j_E (j_B) over a spacelike slice N is the total electric (magnetic) charge. We then have a de Rham cohomology interpretation of the electric (magnetic) charge, $\overline{Q}_E = [j_E|_N]_{dR} \in H^3_c(N;\mathbb{R})$ ($\overline{Q}_B = [j_B|_N]_{dR} \in H^3_c(N;\mathbb{R})$), where the subscript c indicates that the cohomology is taken with compact support. In the quantum theory, much as in the descriptions earlier in this paper, the Dirac quantization condition states that the electric (magnetic) charge is a cohomology class that lies in the image of the map $H^3_c(N;\mathbb{Z}) \to H^3_c(N;\mathbb{R})$. This is useful to generalize to higher gauge fields. When we work in arbitrary dimensions, allowing N to be any oriented Riemannian manifold of dimension $n-1$ (and X the n-dimensional spacetime), then for p-form gauge fields, i.e. for F a homogeneous form of degree $p+1$, so that $\deg j_B = p+2$ and $\deg j_E = n-p$, the pair (Q_E, Q_B) lives in $H^{p+2}(X;\mathbb{Z}) \oplus H^{n-p}(X;\mathbb{Z})$ [Fre02].

While ordinary cohomology can encode the integer magnetic charge, it is insufficient for capturing the induced electric charge shift because that shift depends on a differential refinement of the gauge field. The natural language is that of differential cohomology (see appendix). In differential cohomology, a $U(1)$ gauge field is a differential cocycle $\check{A} \in \check{C}^2(M)$ with $\mathrm{curv}(\check{A}) = \frac{F}{2\pi}$ and $c(\check{A}) = a \in H^2(M;\mathbb{Z})$. If there is a magnetic current, $\check{A}$ is not a cocycle, but a trivialization of a magnetic-current cocycle $d\check{A} = \check{j}_B$. The gauge field is a cochain trivializing the magnetic current. Locally, on a patch $U(i)$, $\check{A} \sim A_i \in \Omega^1(U_i)$, and $F|_{U_i} = dA_i$. When we vary the differential cocycle $\check{A}$, the variation of its curvature is an exact global form in a fixed topological sector. So varying the Lorentzian action with respect to A, using $F = dA$ locally, so $\delta F = d(\delta A)$,

$$S[A] = -\frac{1}{2\alpha^2}\int_M F \wedge \star F + \frac{\theta}{8\pi^2}\int_M F \wedge F$$

$$\delta\left(-\frac{1}{2\alpha^2}\int F \wedge \star F\right) = -\frac{1}{\alpha^2}\int \delta F \wedge \star F = -\frac{1}{\alpha^2}\int d(\delta A) \wedge \star F$$

$$\delta\left(\frac{\theta}{8\pi^2}\int F \wedge F\right) = \frac{\theta}{4\pi^2}\int \delta F \wedge F = \frac{\theta}{4\pi^2}\int d(\delta A) \wedge F$$

Integrating by parts, $-\int d(\delta A) \wedge \star F = \int \delta A \wedge d(\star F)$ up to boundary terms, and $\int d(\delta A) \wedge F = -\int \delta A \wedge dF$ up to boundary terms, so

$$\delta S = \int \delta A \wedge \left[\frac{1}{\alpha^2} d(\star F) - \frac{\theta}{4\pi^2} dF\right]$$

For arbitrary δA, the Euler-Lagrange equation is $\frac{1}{\alpha^2} d(\star F) - \frac{\theta}{4\pi^2} dF = 0$.

If a magnetic current is present, $dF = 2\pi j_B$, so $d\left(\frac{1}{\alpha^2} \star F\right) = \frac{\theta}{2\pi} j_B$. By Maxwell's equation $d\left(\frac{1}{\alpha^2} \star F\right) = j_E$, we get the Witten effect $\Delta j_E = \frac{\theta}{2\pi} j_B$.

7 Conclusion

In the elementary derivation of the Witten effect, it appears as a shift in the electric charge lattice of monopole sectors by an amount proportional to θ. In the cohomological formulation, the same shift is a consequence of the integral structure carried by the gauge field and of the topological pairing encoded by the θ term. This perspective is useful because it separates the local differential data of the gauge field from its global topological content. Ordinary cohomology records quantized magnetic charge, while differential cohomology packages together the curvature, characteristic class, and holonomy data needed to describe the gauge field itself. From this point of view, the Witten effect of four-dimensional electromagnetism is a basic example of how topological terms modify charge quantization in a differential refinement of cohomology.

This suggests a natural direction for higher generalizations. For higher-form gauge fields, electric and magnetic charges are again classified by integral cohomology classes, while the gauge fields themselves are more naturally described by differential cocycles. Topological pairings among these cocycles can similarly shift the allowed electric charge sectors in the presence of magnetic defects. The usual Witten effect is thus a basic four-dimensional example of a broader cohomological mechanism in higher gauge theory.

Appendix A: Differential cohomology

For the sake of brevity, we only give the tools strictly necessary for this paper. The reference is [ADH21].

Differential cohomology theories are sheaves of spectra on the category of manifolds. For M a manifold and $i \geq 0$ an integer, we write $C_i^{sm}(M;\mathbb{Z})$ for the abelian group of smooth, integer-valued chains on M, and $Z_i^{sm}(M;\mathbb{Z}) \subset C_i^{sm}(M;\mathbb{Z})$ for the subgroup of smooth cycles. For $k \geq 1$, a degree k differential character on M is a homomorphism $\chi : Z_{k-1}^{sm}(M;\mathbb{Z}) \to \mathbb{R}/\mathbb{Z}$ such that there exists a k-form $\omega(\chi) \in \Omega^k(M)$ with the property that for every $c \in C_k^{sm}(M;\mathbb{Z})$, $\chi(\partial c) = \int_c \omega(\chi) \mod \mathbb{Z}$. We write

$$\check{H}^k(M;\mathbb{Z}) \subset \mathrm{Hom}_{\mathbb{Z}}(Z_{k-1}^{sm}(M;\mathbb{Z}), \mathbb{R}/\mathbb{Z})$$

for the abelian group of degree k differential characters on M.

There are two natural pieces of data associated to a differential character. First, by definition, every differential character $\chi \in \check{H}^k(M;\mathbb{Z})$ has a curvature form $\mathrm{curv}(\chi) = \omega(\chi) \in \Omega^k(M)$. The defining condition implies that $\omega(\chi)$ is closed and has integral periods, i.e. $\int_z \omega(\chi) \in \mathbb{Z}$ for every smooth k-cycle z. Thus the curvature map has image

$$\Omega^k_{\mathbb{Z}}(M) := \{\omega \in \Omega^k(M) : d\omega = 0, \int_z \omega \in \mathbb{Z} \text{ for all } z \in Z_k^{sm}(M;\mathbb{Z})\}.$$

We therefore have a natural homomorphism $\mathrm{curv} : \check{H}^k(M;\mathbb{Z}) \longrightarrow \Omega^k_{\mathbb{Z}}(M)$. The kernel of this map consists of flat differential characters. These are precisely ordinary $\mathbb{R}/\mathbb{Z}$-valued cohomology classes of degree $k-1$. Hence there is a short exact sequence

$$0 \longrightarrow H^{k-1}(M;\mathbb{R}/\mathbb{Z}) \longrightarrow \check{H}^k(M;\mathbb{Z}) \xrightarrow{\mathrm{curv}} \Omega^k_{\mathbb{Z}}(M) \longrightarrow 0.$$

In this sense, differential cohomology refines closed differential forms with integral periods by remembering their possible flat holonomy data.

There is also a characteristic-class map $c : \check{H}^k(M;\mathbb{Z}) \longrightarrow H^k(M;\mathbb{Z})$ which one can define as follows. If χ is represented by a differential cocycle (c, h, ω), then c is an integral cocycle and its cohomology class $[c] \in H^k(M;\mathbb{Z})$ depends only on the differential cohomology class of χ. We define $c(\chi) := [c]$. The image of $c(\chi)$ in real cohomology is represented by the curvature form:$c(\chi) \otimes 1 = [\mathrm{curv}(\chi)] \in H^k(M;\mathbb{R})$. Thus a differential cohomology class simultaneously records three kinds of data: a closed differential form with integral periods, an underlying integral cohomology class, and holonomy information around $(k-1)$-cycles. In the case $k = 2$, this is exactly the data of a $U(1)$ line bundle with connection: the curvature remembers the normalized field strength $F/2\pi$, the characteristic class remembers the first Chern class, and the differential character itself remembers holonomies.

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